

## BELLE ILE-LE TALUT, France

WMO: 072070

Lat: 47.2944N

Lon: 3.2183W

Elev: 43

StdP: 100.81

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 0.1        | 1.4 | -5.0                          | 2.5 | 1.7  | -2.9 | 3.0 | 2.7  | 18.4                  | 10.3 | 16.7 | 9.9  | 6.6                   | 70   | 0.661 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 5.4                    | 25.7            | 19.1 | 23.8 | 18.4 | 22.2 | 17.8 | 20.3                | 23.8 | 19.3 | 22.3 | 18.6 | 21.1 | 4.6                  | 30   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.1                            | 13.9 | 21.5 | 18.2 | 13.2 | 20.5 | 17.6 | 12.7 | 19.9 | 58.3          | 23.9 | 55.2 | 22.5 | 52.9 | 21.1 | 24.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 16.0          | 14.1 | 12.4 | -1.9                       | 29.0 | 2.3                | 1.9 | -3.5                                               | 30.3 | -4.9       | 31.5 | -6.2       | 32.6 | -7.8       | 33.9 |    |    |
| (5)               |      |      | -3.1                       | 21.5 | 2.2                | 1.2 | -4.6                                               | 22.4 | -5.9       | 23.2 | -7.1       | 23.9 | -8.6       | 24.8 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 13.0        | 8.2   | 8.0   | 9.5   | 11.1  | 13.8  | 16.6  | 18.3  | 18.4  | 16.9  | 14.6  | 11.4  | 9.1   |  |  |
|                                                                        | DBStd       | 4.50        | 3.13  | 2.68  | 2.16  | 2.23  | 2.32  | 2.37  | 2.23  | 1.94  | 1.96  | 2.19  | 2.63  | 3.31  |  |  |
|                                                                        | HDD10.0     | 253         | 69    | 62    | 33    | 13    | 1     | 0     | 0     | 0     | 0     | 2     | 17    | 56    |  |  |
|                                                                        | HDD18.3     | 2019        | 315   | 290   | 274   | 218   | 142   | 64    | 27    | 22    | 52    | 117   | 209   | 287   |  |  |
|                                                                        | CDD10.0     | 1348        | 12    | 6     | 17    | 45    | 119   | 197   | 258   | 259   | 206   | 144   | 58    | 27    |  |  |
|                                                                        | CDD18.3     | 72          | 0     | 0     | 0     | 0     | 2     | 11    | 27    | 23    | 8     | 1     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 212         | 0     | 0     | 0     | 0     | 4     | 41    | 91    | 62    | 14    | 0     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 32          | 0     | 0     | 0     | 0     | 0     | 5     | 14    | 11    | 2     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 6.2         | 7.3   | 6.6   | 6.4   | 6.2   | 5.8   | 5.2   | 5.4   | 5.2   | 5.5   | 6.3   | 7.2   | 7.8   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 708         | 87    | 64    | 49    | 51    | 45    | 35    | 40    | 39    | 54    | 77    | 85    | 83    |  |  |
|                                                                        | PrecMax     | 905         | 189   | 133   | 160   | 118   | 105   | 69    | 82    | 99    | 163   | 148   | 191   | 150   |  |  |
|                                                                        | PrecMin     | 473         | 13    | 13    | 2     | 13    | 10    | 4     | 7     | 2     | 2     | 18    | 17    | 19    |  |  |
|                                                                        | PrecStd     | 95          | 49    | 36    | 33    | 30    | 20    | 20    | 21    | 26    | 38    | 34    | 41    | 40    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 13.5  | 12.9  | 17.3  | 19.7  | 23.9  | 27.2  | 28.5  | 28.3  | 25.2  | 20.2  | 16.9  | 14.7  |  |  |
|                                                                        |             | MCWB        | 12.1  | 10.7  | 12.1  | 14.1  | 17.2  | 19.7  | 19.6  | 20.2  | 18.2  | 17.1  | 14.9  | 13.0  |  |  |
|                                                                        | 2%          | DB          | 12.8  | 12.1  | 14.3  | 17.4  | 21.3  | 24.3  | 25.9  | 24.8  | 22.6  | 18.8  | 16.1  | 14.1  |  |  |
|                                                                        |             | MCWB        | 11.7  | 10.7  | 10.7  | 12.9  | 16.4  | 18.2  | 18.9  | 19.4  | 17.3  | 16.6  | 14.5  | 12.7  |  |  |
|                                                                        | 5%          | DB          | 12.3  | 11.6  | 12.9  | 15.7  | 19.0  | 22.2  | 23.6  | 22.7  | 21.2  | 18.1  | 15.4  | 13.6  |  |  |
|                                                                        |             | MCWB        | 11.2  | 10.4  | 10.4  | 12.2  | 15.3  | 17.3  | 18.7  | 18.6  | 17.0  | 16.2  | 13.9  | 12.3  |  |  |
|                                                                        | 10%         | DB          | 11.8  | 11.2  | 12.1  | 14.2  | 17.3  | 20.4  | 21.9  | 21.3  | 19.9  | 17.4  | 14.9  | 13.1  |  |  |
|                                                                        |             | MCWB        | 10.7  | 10.0  | 10.3  | 11.5  | 14.4  | 16.6  | 18.0  | 18.1  | 16.8  | 15.7  | 13.4  | 11.8  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 12.7  | 11.7  | 12.7  | 15.2  | 18.4  | 20.8  | 21.3  | 21.8  | 19.9  | 18.0  | 15.8  | 13.7  |  |  |
|                                                                        |             | MCDB        | 13.0  | 12.0  | 15.5  | 18.5  | 22.4  | 24.7  | 25.1  | 25.8  | 23.1  | 19.1  | 16.4  | 14.1  |  |  |
|                                                                        | 2%          | WB          | 12.1  | 11.2  | 11.7  | 13.6  | 16.9  | 19.1  | 20.1  | 20.3  | 18.7  | 17.3  | 15.0  | 13.2  |  |  |
|                                                                        |             | MCDB        | 12.6  | 11.6  | 12.9  | 16.2  | 20.0  | 22.8  | 23.9  | 23.4  | 20.9  | 18.2  | 15.6  | 13.7  |  |  |
|                                                                        | 5%          | WB          | 11.5  | 10.8  | 11.2  | 12.7  | 15.8  | 18.1  | 19.2  | 19.3  | 18.0  | 16.7  | 14.3  | 12.7  |  |  |
|                                                                        |             | MCDB        | 12.1  | 11.3  | 12.2  | 14.8  | 18.5  | 21.2  | 22.4  | 21.7  | 19.9  | 17.7  | 15.2  | 13.4  |  |  |
|                                                                        | 10%         | WB          | 10.9  | 10.4  | 10.8  | 12.0  | 14.8  | 17.1  | 18.5  | 18.6  | 17.4  | 16.1  | 13.6  | 12.2  |  |  |
|                                                                        |             | MCDB        | 11.6  | 11.0  | 11.7  | 13.6  | 16.8  | 19.7  | 21.2  | 20.6  | 19.2  | 17.1  | 14.8  | 12.9  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 3.2   | 3.7   | 4.3   | 5.0   | 5.0   | 5.4   | 5.3   | 5.4   | 5.4   | 3.9   | 3.5   | 3.3   |  |  |
|                                                                        |             | MCDBR       | 2.6   | 3.2   | 5.5   | 7.4   | 7.9   | 8.7   | 9.0   | 7.8   | 7.2   | 4.1   | 3.0   | 2.5   |  |  |
|                                                                        | 5% WB       | MCWBR       | 2.9   | 3.1   | 3.6   | 4.4   | 4.5   | 4.6   | 4.5   | 3.9   | 3.6   | 2.9   | 3.1   | 2.9   |  |  |
|                                                                        |             | MCWBR       | 2.5   | 2.8   | 4.0   | 5.6   | 6.8   | 7.2   | 7.1   | 6.2   | 5.1   | 3.2   | 2.7   | 2.4   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.337 | 0.348 | 0.374 | 0.387 | 0.392 | 0.391 | 0.387 | 0.377 | 0.364 | 0.364 | 0.342 | 0.330 |  |  |
|                                                                        |             | taud        | 2.433 | 2.420 | 2.366 | 2.349 | 2.382 | 2.413 | 2.425 | 2.469 | 2.484 | 2.471 | 2.450 | 2.412 |  |  |
|                                                                        | Ebn at Noon |             | 740   | 813   | 843   | 867   | 873   | 873   | 872   | 868   | 847   | 785   | 726   | 697   |  |  |
|                                                                        |             | Edh at Noon | 69    | 85    | 104   | 115   | 116   | 113   | 110   | 101   | 91    | 80    | 67    | 63    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.09  | 2.04  | 3.28  | 4.89  | 5.78  | 6.33  | 6.06  | 5.31  | 4.22  | 2.41  | 1.38  | 0.95  |  |  |
|                                                                        | RadStd      |             | 0.10  | 0.20  | 0.33  | 0.46  | 0.38  | 0.54  | 0.26  | 0.35  | 0.28  | 0.20  | 0.12  | 0.10  |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (26 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page